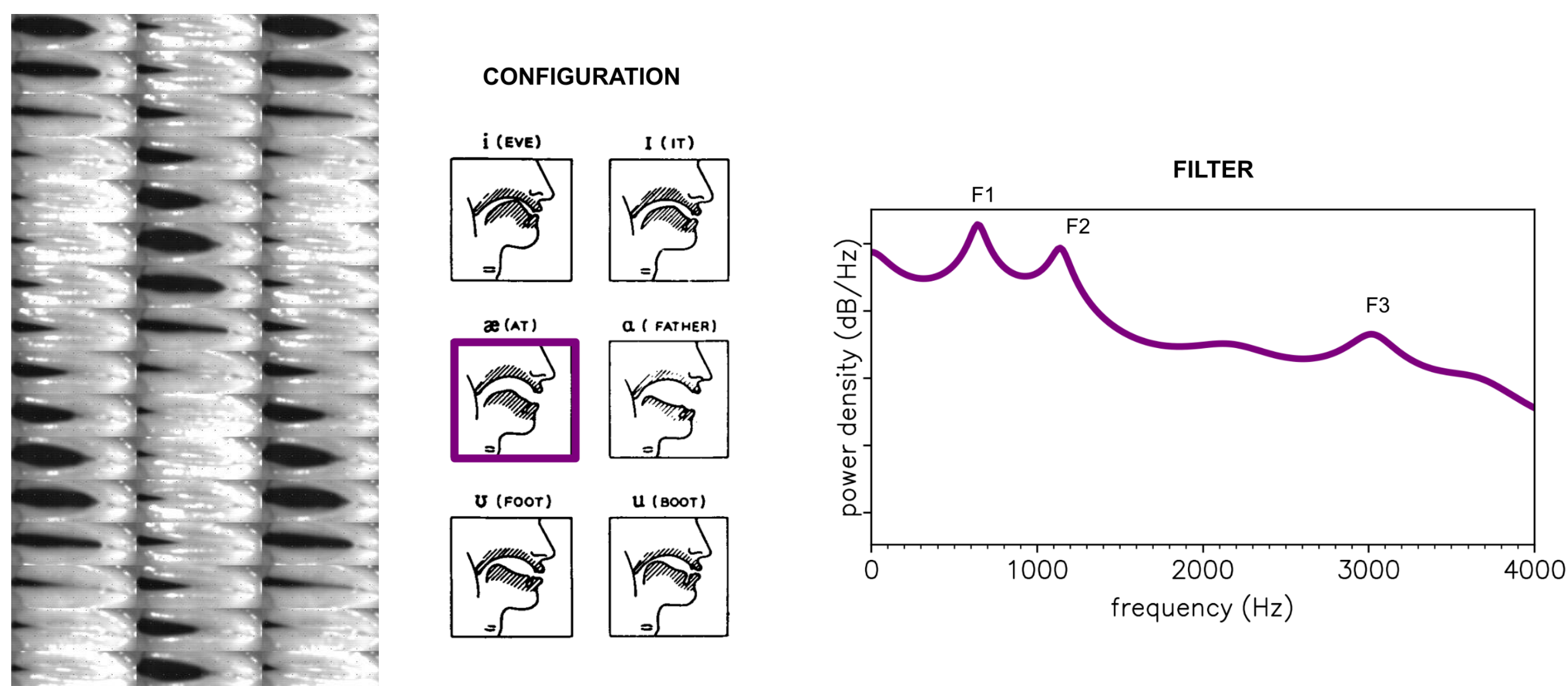


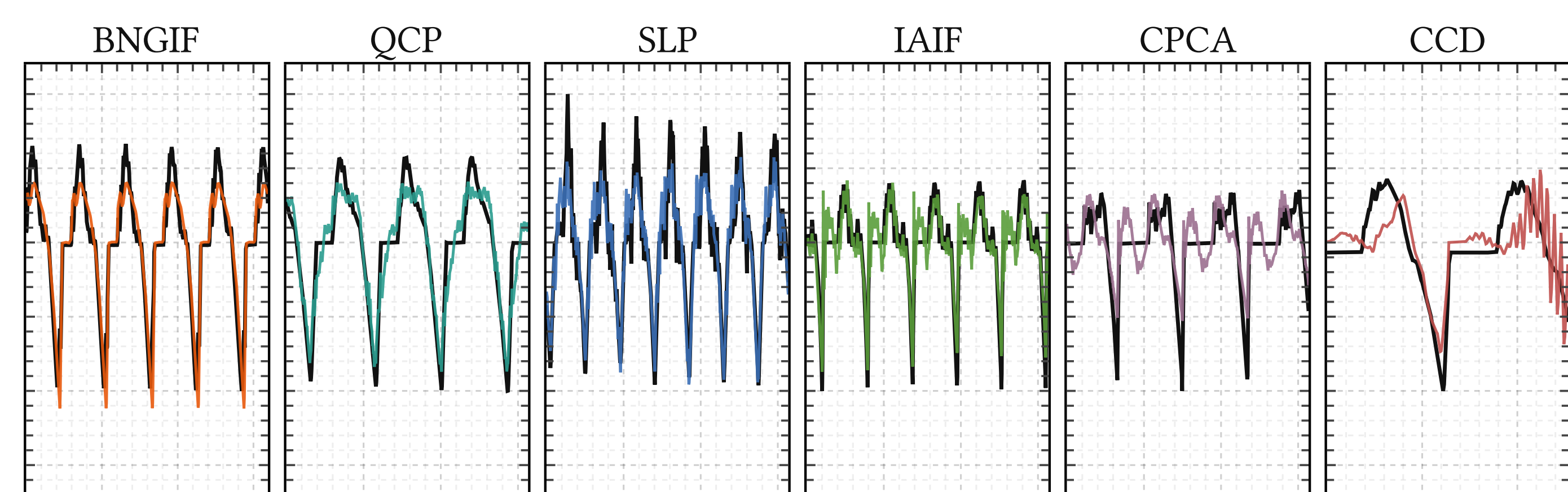
PROBABILISTIC SPEECH DECONVOLUTION IN REAL TIME

SPEECH = SOURCE × FILTER

The vocal fold airflow and the vocal tract, from the microphone alone.



BEATS THE STATE OF THE ART



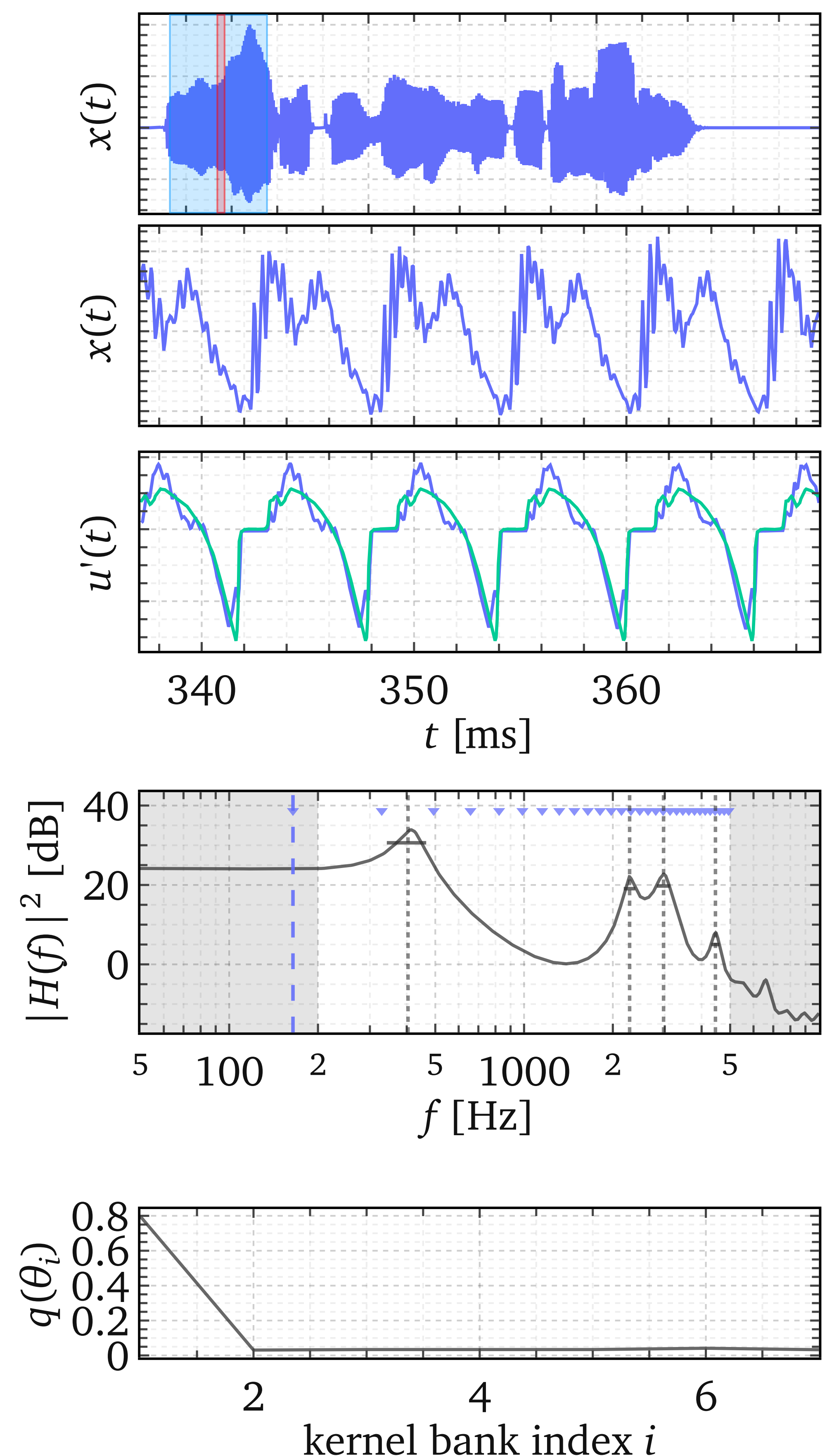
True airflow (black) and six methods' estimates, one frame of the EGIFA benchmark.

	BNGIF	QCP	SLP	IAIF	CPCA	CCD
BNGIF	0.935	<.001	<.001	<.001	<.001	<.001
QCP	1.000	0.910	<.001	<.001	<.001	<.001
SLP	1.000	1.000	0.889	<.001	<.001	<.001
IAIF	1.000	1.000	1.000	0.886	<.001	<.001
CPCA	1.000	1.000	1.000	1.000	0.759	0.004
CCD	1.000	1.000	1.000	1.000	0.996	0.741

Over the whole benchmark: better than each of them, $p < .001$.

ONE MODEL, ALL THE UNCERTAINTY

Posterior over the airflow (band), the filter, and which prior shape explains the frame.



5× REAL TIME ON ONE GPU

Variational inference in JAX: a 32 ms frame in 6 ms.

WHY IT MATTERS

Hearing implants – pitch and formants with error bars.

Clinics – physiological voice biomarkers, not black-box features.

Phonetics, forensics – formant tracks 30% more accurate than Praat.

